Todo list

Missing dissertation outline 6

Missing turbulence overview 9

Missing optics overview 9

Missing hypersonic modeling 10

Missing turbulence modeling 10

Missing comparison between equilibrium and nonequilibrium 11

Contents

1 Introduction **3**

1.1 Motivation 3

1.2 Dissertation Outline 6

2 Theory **7**

2.1 Hypersonic Overview 7

2.2 Turbulence Overview 9

2.3 Optics Overview 9

3 Modeling **10**

3.1 Hypersonic Modeling 10

3.2 Turbulence Modeling 10

4 Numerical Results **11**

4.1 Ionization Study 11

Abbreviations **13**

Symbols **14**

Appendices **16**

Chapter 1

Introduction

1.1 Motivation

Several reports in the classified and unclassified [1], [2] environment have stated the concern of aero-optic effects at hypersonic speeds to be of paramount importance for national security. Applied work in this field is not only being done by national laboratories such as Lawrence Livermore National Laboratory (LLNL), Applied Physics Laboratory (APL), Los Alamos National Laboratory (LANL) but also by the Department of Defense (DoD) contractors such as Lockheed Martin Corp., Northrop Grumman Corp., Raytheon Technologies. As expected most of this work is limited in the open literature. However, work that investigates the effect of optical properties in a hypersonic environment is lacking and there are many fundamental phenomenons such as index of refraction, that needs to be better understood.

Understanding the effects of the environment in which an optical beam traverse is imperative in target detection. Radar engineers will be mainly concerned with the atmospheric environment (rain, clouds, fog, snow) but will not much much consideration on the impact aerodynamic effects such as shock wave and boundary layer interactions has on the optical beam. Some work showing how much the index of refraction is affected by aerodynamic effects has been performed bu Gupta et.hal. [3]

for supersonic flows and showed that there is indeed, a change on index of refraction across a shock wave. For subsonic flows, it might be a valid assumption to ignore the aerodynamic effects on the optical beam; however for supersonic and specially for hypersonic flows this assumption has to be revisit. During a hypersonic flight the temperatures behind the shock are very high, that will no only induce thermal noise on the sensor but will also affect the electromagnetic properties that will influence the sensor's performance. Hence a higher distortion as well as power loss will be expected in a hypersonic flow. Being able to accurately understand this is of importance to build accurate sensor that can not only target objects moving at these speeds, but can also accurate predict targets from an object moving at hypersonic speeds.

Fig.1.1, shows the path that a beam in a hypersonic flight will follow. The beam has to travel across the isolated gas, the radome, the boundary layer, shock layer and lastly environment. Traditionally, radar engineers ignore aerodynamic effects (shock waves, expansion waves) for their analysis; however for the case of hypersonic flights the aerodynamic effects cannot be ignored anymore, because the electromagnetic properties across the boundary layer and shock layer change.

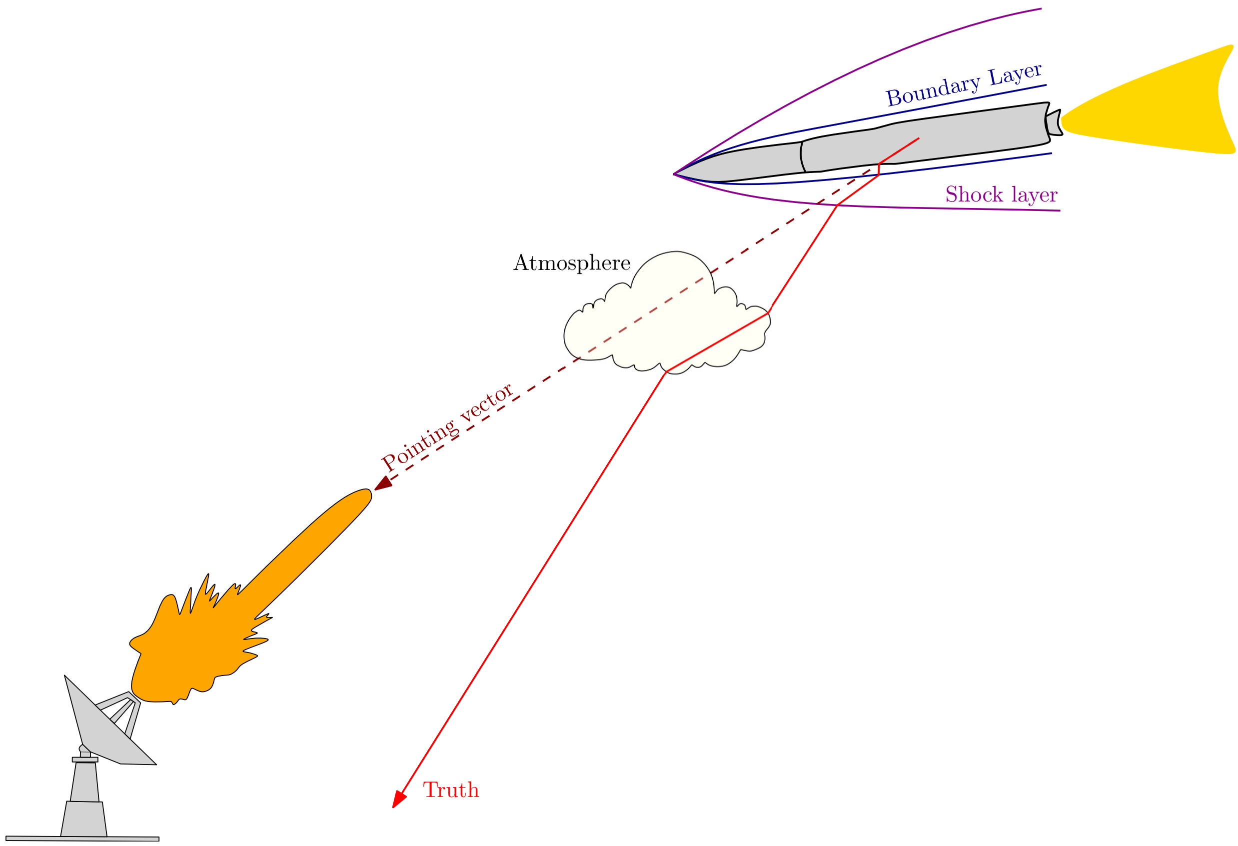

Figure 1.1: Optics Schematic

Aero-optical investigations can be split into four quadrants as shown in Tab.1.1. The optical parameters are contingent on the degree to which turbulence eddy resolution and turbulence fluctuations are captured in the Computational Fluid Dynamics (CFD) simulation. In cases where a lower level of flow resolution is utilized, either a laminar flow or a Reynolds Averaged Navier-Stokes (RANS) simulation are employed as shown in Quadrants I and III of Tab.1.1. While a RANS approach averages out turbulent fluctuations, it is a popular approach for hypersonic vehicle analysis due to it's efficiency [4] and can still provide aero-optical quantities such as Gladstone-Dale constant, polarizability, Bore-Sight Error (BSE), Optical Path Lenght (OPL), and Wavefront Distortion (WD). The amount of enthalpy is also an important criteria in the flow field. With high enthalpy flows, many physical phenomena in the flow field that are not present in low-speed flows present themselves. This includes nonequilibrium, reactive flows, and excitation of molecular internal modes, all of which can affect optical signals either directly or indirectly; these are shown in the top Quadrants.

This work aims to investigate and compare all quadrants using different fidelities of nonequilibrium and fluid flows.

Quadrant III: High-Enthalpy, Low-Flow Resolution *Physics captured:* chemical and thermal nonequilibrium, shock structures, bulk unsteadiness, bulk density gradients. *AO quantities measured:* Gladstone-Dale Constant, Polarizability, Dielectric Constant, BSE, OPL, WD.	**Quadrant IV:** High-Enthalpy, High-Flow Resolution *Physics captured:* chemical and thermal nonequilibrium, shock structures, bulk unsteadiness, bulk density gradients. *AO quantities measured:* Gladstone-Dale Constant, Polarizability, Dielectric Constant, BSE, OPL, WD, OPD.
Quadrant I: Low-Enthalpy, Low-Flow Resolution *Physics captured:* shock structures, bulk unsteadiness, bulk density gradients. *AO quantities measured:* Gladstone-Dale Constant, Polarizability, Dielectric Constant, BSE, OPL, WD.	**Quadrant II:** Low-Enthalpy, High-Flow Resolution *Physics captured:* shock structures, bulk unsteadiness, bulk density gradients. *AO quantities measured:* Gladstone-Dale Constant, Polarizability, Dielectric Constant, BSE, OPL, WD, OPD.

Table 1.1: Schematic of the fidelity of aero-optical simulations.

A conference paper focusing on quadrants I and III was published by Liza et al. [5]. In this paper, the authors examined the impact of high temperatures and nonequilibrium conditions on polarizability. Additionally, they investigated the effects of Mach number, turbulence, and chemistry modeling on optical properties, including the index of refraction and the dielectric constant of air.

1.2 Dissertation Outline

Missing dissertation outline

Chapter 2

Theory

2.1 Hypersonic Overview

The first hypersonic flight took place during the early stages of the Cold War, in 1949 when the German V2 rocket developed during the Second World War in Nazi Germany and brought to the USA (possible during operation Paperclip) reached hypersonic speeds as it entered the atmosphere [6]. In 1961, the Soviet Union cosmonaut Yuri Garin became the fist person to safely reach space and orbit Earth. During his journey he became the first human in history to travel at hypersonic speeds [6].

Speeds in a hypersonic flow are greater than Mach five, at these speeds the kinetic energy of the gas molecules is converted into thermal energy, leading to very high temperatures. At these speeds the chemical composition of the gas becomes important. The high temperature can cause dissociation (breaking apart of molecules), and ionization (formation of charged particles) of the air molecules. At temperatures above 2000 $[K]$ O_2 starts to dissociate, and it is completely dissociated above 4000 $[K]$. N_2 begins to dissociate above 4000 $[K]$ and it is completely dissociated above 9000 $[K]$ [7].

At these temperatures, the Hamiltonian (kinetic and potential energy) of the system has to be modified because more energy modes are accessible (rotational, vibrational, and electronic energy). Fig.2.1 shows the additional degrees of freedom that emerge during a hypersonic flow.

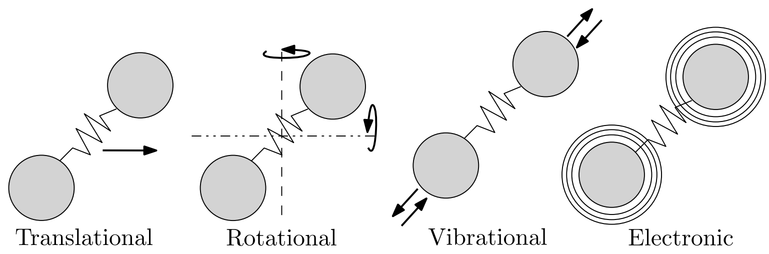

Figure 2.1: Additional Degrees of Freedom

In addition, each new accessible energy mode acts at different time scales, Eq.(2.1) show these time scales.

$$\tau_{flow} \approx \tau_{tr} < \tau_{rot} < \tau_{vib} < \tau_{chem} \tag{2.1}$$

The different time scales is the reason why flows at hypersonic speeds are in nonequilibrium. After air molecules travel across a shock, it will take certain number of collisions to convert the energy into thermal energy. In the case of translational energy is almost instantaneous, in the case of rotational is approximately 10 collisions and in the case of vibrational is approximately 1000 collisions [8]. The nonequilibrium characteristics of the flow can be characterize by comparing the flow characteristic time with the other time scales [9].

These additional energy modes increase the degrees of freedom (dof) of the system, allowing molecules to occupy more energy levels (rotational, vibrational, electronic). Due to this, the partition function (which indicates how particles in an assembly are partitioned among various energy levels of an atom or molecule [10]) has to be modified to consider these additional dofs. In statistical mechanics, the partition function is used to calculate the heat capacities of the system, which are then used to calculate macro properties such as heat transfer and temperature, which are employed in fluid dynamics.

At these conditions, the calorically perfect gas approximation is no longer suitable because this assumption does not consider thermochemical effects. The calorically perfect gas assumption will lead to errors such as over-predicting temperature, under-predicting density, and underestimating pressure. Therefore, at these speeds, thermochemical effects need to be considered and accurate models that consider these effects need to be included.

2.2 Turbulence Overview

Missing turbulence overview

2.3 Optics Overview

Missing optics overview

Chapter 3

Modeling

3.1 Hypersonic Modeling

Missing hypersonic modeling

3.2 Turbulence Modeling

Missing turbulence modeling

Chapter 4

Numerical Results

4.1 Ionization Study

This investigation compares the index of refraction between a 5 and 11 species gas. The solver used for this investigation is Le Michigan Aerothermodynamic Navier-Stokes Solver (LeMANS), which is a reactive hypersonic flow code [11]. Tab.4.1 shows the cases that were tested; the simulations described below use a isothermal BC, with the famous Park's $2T$ [12] model.

Case #	Geometry	# of Species	AoA $[\circ]$	M [ ]	Alt $[km]$
1	flat plate	5	10	15	30
2	flat plate	11	10	15	30

Table 4.1: Ionization table study

The equilibrium polarizability constants (neutral) and the nonequilibrium polarizability constants (ions) used for this study are provided on Tab.4.2 and Tab.4.3.

Species	$\alpha \times 10^{30}$ $[m]$	Ref.
N_2	1.740	[13]
N	1.100	[13]
O_2	1.581	[13]
O	0.802	[13]
NO	1.700	[13]

Table 4.2: Equilibrium polarizability constants for ions and neutral species

Species	$\alpha \times 10^{30}$ $[m]$	Ref.
N_2^+	2.386	[14]
N^+	0.559	[15]
O_2^+	0.238	[16]
O^+	0.345	[16]
NO^+	1.021	[17]

Table 4.3: Nonequilibrium polarizability constants for ions and neutral species

Missing comparison between equilibrium and nonequilibrium

Bibliography

[1] C. H. Jones, "Recommendations from the workshop on communications through plasma during hypersonic flight," Tech. Rep., 2009. DOI: 10.2514/6.2009-1718.

[2] Chief Scientist (Air Force), "Technology Horizons A Vision for Air Force Science and Technology 2010-30," United States Air Force Chief Scientist, Tech. Rep., 2010.

[3] A. Gupta and B. Argrow, "Analytical Approach for Aero-Optical and Atmospheric Effects in Supersonic Flow Fields," in *AIAA Scitech 2020 Forum*, Reston, Virginia: American Institute of Aeronautics and Astronautics, Jan. 2020, ISBN: 978-1-62410-595-1. DOI: 10.2514/6.2020-0684. [Online]. Available: https://arc.aiaa.org/doi/10.2514/6.2020-0684.

[4] G. V. Candler, "Rate Effects in Hypersonic Flows," *Annual Review of Fluid Mechanics*, vol. 51, pp. 379–402, 2019, ISSN: 00664189. DOI: 10.1146/annurev-fluid-010518-040258.

[5] M. Liza, O. Tumuklu, and K. M. Hanquist, "Nonequilibrium Effects on Aero-Optics in Hypersonic Flows," in *AIAA AVIATION 2023 Forum*, Reston, Virginia: American Institute of Aeronautics and Astronautics, Jun. 2023, ISBN: 978-1-62410-704-7. DOI: 10.2514/6.2023-3736. [Online]. Available: https://arc.aiaa.org/doi/10.2514/6.2023-3736.

[6] J. Anderson and B. Mary, *Introduction to Flight*, 9th. McGraw-Hill Higher Education, 2022, ISBN: 1260226743.

[7] J. D. Anderson, *Hypersonic and High Temperature Gas Dynamics*, 2nd. American Institute of Aeronautics and Astronautics, 2006, p. 811, ISBN: 1563477807, 9781563477805.

[8] C. Vincenti and W. Kruger, *Introduction to Physical Gas Dynamics*. Krieger Pub Co., Jul. 1975, ISBN: 0882753096.

[9] K. J. Neitzel, "Thermochemical Modeling of Nonequilibrium Oxygen Flows," Ph.D. dissertation, The University of Mikchigan, 2017. [Online]. Available: https://deepblue.lib.umich.edu/bitstream/handle/2027.42/137049/kneitzel_1.pdf?sequence=1.

[10] N. M. Laurendeau, *Statistical Thermodynamics, Fundamentals and Aplications*. Cambridge University Press, 2005, ISBN: 9780521846356.

[11] L. C. Scalabrin, "Numerical Simulation of Weakly Ionized Hypersonic Flow over Reentry Capsules," Ph.D. dissertation, University of Michigan, 2007.

[12] C. Park, "Assessment of two-temperature kinetic model for ionizing air," *Journal of Thermophysics and Heat Transfer*, vol. 3, no. 3, pp. 233–244, 1989, ISSN: 15336808. DOI: 10.2514/3.28771.

[13] W. Haynes, *CRC Handbok of Chemistry and Physics*, 95th. 2004, ISBN: 978-1-4822-0868-9.

[14] E. F. McCormack, S. T. Pratt, J. L. Dehmer, and P. M. Dehmer, "Analysis of the 8 f , 9 f , and 10 f , v =1 Rydberg states of N 2," *Physical Review A*, vol. 44, no. 5, pp. 3007–3015, Sep. 1991, ISSN: 1050-2947. DOI: 10.1103/PhysRevA.44.3007. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevA.44.3007.

[15] P. L. Jacobson, R. D. Labelle, W. G. Sturrus, R. F. Ward, and S. R. Lundeen, "Optical spectroscopy of high- L n =10 Rydberg states of nitrogen," *Physical Review A*, vol. 54, no. 1, pp. 314–322, Jul. 1996, ISSN: 1050-2947. DOI: 10.1103/PhysRevA.54.314. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevA.54.314.

[16] R. Stewart, "A numerical study of coupled Hartree-Fock theory for open-shell systems," *Molecular Physics*, vol. 30, no. 4, pp. 1283–1288, Oct. 1975, ISSN: 0026-8976. DOI: 10.1080/00268977500102811. [Online]. Available: http://www.tandfonline.com/doi/abs/10.1080/00268977500102811.

[17] M. Fehér and P. Martin, "Ab initio calculations of the properties of NO + in its ground electronic state X 1 Σ +," *Chemical Physics Letters*, vol. 215, no. 6, pp. 565–570, Dec. 1993, ISSN: 00092614. DOI: 10.1016/0009-2614(93)89356-M. [Online]. Available: https://linkinghub.elsevier.com/retrieve/pii/000926149389356M.

Abbreviations

N-S Navier-Stokes

LeMANS Le Michigan Aerothermodynamic Navier-Stokes Solver

CFD Computational Fluid Dynamics

RANS Reynolds Averaged Navier-Stokes

LES Large Eddy Simulations

DNS Direct Numerical Simulation

URANS Unsteady Reynolds Averaged Navier-Stokes

DES Detached Eddy Simulation

DoD Department of Defense

DoE Department of Energy

LLNL Lawrence Livermore National Laboratory

LANL Los Alamos National Laboratory

APL Applied Physics Laboratory

BSE Bore-Sight Error

OPL Optical Path Lenght

WD Wavefront Distortion

dof degrees of freedom

Symbols

Re	Reynolds Number
ρ	density $\left[\frac{M}{L^3}\right]$
μ	dynamic viscosity $\left[\frac{M}{LT}\right]$
ν	kinematic viscosity $\left[\frac{L^2}{T}\right]$
L_c	characteristic length
p	pressure
κ	thermal conductivity
E	total energy
τ_{ij}	shear stress tensor $\left[\frac{M}{LT^2}\right]$
S_{ij}	strain-rate tensor $\left[\frac{1}{T}\right]$
Ω_{ij}	rotation-rate tensor
f_i	body force
ϵ	dissipation rate
η	Kolmogorov length scale
τ	Kolmogorov time scale
υ	Kolmogorov velocity scale
$f(r,t)$	Longitudinal velocity autocorrelation function
$L_{11}(t)$	Longitudinal integral scale
$\lambda_f(t)$	Longitudinal Taylor microscale
$g(r,t)$	Transverse velocity autocorrelation function
$L_{22}(t)$	Transverse integral scale

$\lambda_g(t)$	Transverse Taylor microscale
δ	velocity boundary layer thickness
δ_T	thermal boundary layer thickness
Pr	Prandlt Number
$(\ )^R$	resolved scale quantity
$(\ )^M$	modeled scale quantity (subgrid scale quantity)
$\overline{(\ )}$	mean quantity
$(\ )'$	fluctuation quantity
$(\ )_{DNS}$	quantitiy from DNS
M_t	turbulent mach number
a	speed of sound
k	kinetic energy
u^{sol}	solenoidal field
u^{dil}	dilatational field

Appendices